

UNIREFLOW: UNIFIED ONE-STEP PERCEPTION AND GENERATION VIA RECTIFIED FLOW DISTILLATION

¹Muhammad Ridha Agam (阿杠), ²Novendra Setyawan (諾文德拉), ³Chi-Chia Sun (宋啟嘉), ⁴Hui-Kai Su (蘇暉凱), ⁴Jun-Wei Hsieh (謝君偉)

¹ Department of Electrical Engineering, National Formosa University, Taiwan

² Graduate Institute of Electro-Optical Engineering, National Formosa University, Taiwan

³ Department of Computer Science and Information Engineering, National Taipei University, Taiwan

⁴ Institute of Computational Intelligence, National Yang Ming Chiao Tung University, Taiwan

E-mail: 11365165@nfu.edu.tw

ABSTRACT

Semantic segmentation and semantic image synthesis are dual tasks in visual perception and generation. Recent unified flow models treat them as a single ODE transporting between image and label map distributions, but require 25 integration steps, which limits throughput. We propose UniReflow, the first framework to run both tasks in a single step. We observe that image and label map latents share spatial structure: boundaries, layout, and semantic extent. This keeps the transport close to linear, which we confirm by measuring trajectory straightness directly. A single rectified-flow reflow iteration is then enough to straighten the trajectories for one-step Euler integration, where noise-to-image distillation needs several. We distill the model with a perceptual loss for synthesis and MSE for segmentation. On COCO-Stuff and CelebAMask-HQ, our one-step UniReflow closely matches its 25-step teacher in a single forward pass, making unified perception and generation practical on consumer GPUs.

Keywords: *rectified flow distillation; one-step inference; semantic segmentation; semantic image synthesis; unified perception and generation.*

1. INTRODUCTION

Semantic segmentation (SS) and semantic image synthesis (SIS) form a natural pair: SS maps an image to a per-pixel label map, and SIS reverses the direction, generating a photorealistic image from a label map. Recent unified flow models exploit this duality by treating the two tasks as a single ordinary differential equation (ODE) between the image and mask latent distributions, with the direction of ODE integration determining the task [1]. Prior unified flow models such as SemFlow [1] require 25 Euler steps per inference, which limits throughput on consumer hardware.

Rectified flow [2] and reflow distillation [3] can in principle collapse the multi-step inference of a flow model into a single forward pass, and a fast read of our work would call it “SemFlow plus InstaFlow.” That misses two things. First, SemFlow [1] stops at the 25-step teacher and

never distills to one step; we are the first to operationalise one-step inference for unified image-mask transport and to measure why a single reflow iteration is enough. Second, InstaFlow [3] needs two full reflow rounds for competitive one-step quality and adapts attention layers only. For image-mask transport one reflow round suffices, but the adapters must also cover the residual-block convolutions that carry sharp boundaries (Sec. 3.2), so the noise-to-image recipe does not transfer as-is. Our contribution is therefore a single direction-conditioned UNet that does both tasks in one Euler step after one reflow iteration, with a transport-geometry explanation backed by direct measurement.

A learnable direction embedding, added to the diffusion time embedding via the UNet class-embedding slot, routes the same network to either the forward (mask) or reverse (image) output, and per-direction LoRA adapters of identical rank specialise the backbone for each task: MSE on segmentation reflow pairs and LPIPS [25] on synthesis reflow pairs. Image and mask latents share boundaries, layout, and semantic extent, so the underlying transport is close to linear, which is why a single reflow iteration suffices here and would not in the standard noise-to-image setting [3].

Past work argues the near-linearity of image-mask transport through geometric intuition alone. We compute the trajectory straightness integral on two datasets and against an image-noise baseline, showing that paired-modality transport is empirically straighter by more than an order of magnitude. The same teacher evaluated on (image, mask) pairs is between 11.3× and 38.4× straighter than when evaluated on (image, noise) pairs, which turns the intuition into evidence and explains why one reflow iteration is sufficient in our setting.

Contributions.

(1) One-step image-mask transport is operational, not a lossy shortcut. A single Euler step after one reflow iteration recovers the 25-step teacher within 1.6 mIoU on COCO-Stuff SS, and the same teacher at one step without reflow collapses (Table 3a). This inverts the one-step-as-compromise framing standard in noise-to-image distillation.

(2) Trajectory near-linearity is measured, not argued. On the same teacher and the same data, the paired (image, mask) coupling reduces the straightness integral S by $11.3\times$ on COCO-Stuff and $38.4\times$ on CelebAMask-HQ relative to the unpaired (image, noise) coupling (Table 2). This is a controlled measurement of how much endpoint coupling alone bends the trajectory.

(3) One unified architecture, one forward pass per task. A single direction-conditioned UNet routed by a learnable direction embedding, with per-direction LoRA adapters of identical rank, runs each task in one forward pass at inference. Replacing 25 Euler steps with one holds the teacher’s mIoU and FID on both benchmarks, so the unified flow keeps its full quality at a fraction of the inference cost.

2. RELATED WORK

Unified perception and generation. SemFlow [1] formulated SS and SIS as opposing ODE directions of a single rectified flow trained on image-mask pairs. FreestyleNet [18] applies rectified flows to layout-to-image generation but is unidirectional. Diffusion-based unifiers such as BBDM [4], Palette [19], and PITI [20] frame translation as denoising but require dozens of steps; ControlNet [21] adds spatial conditioning to pre-trained latent diffusion [9] but does not unify perception. None of these works deliver a single one-step deployment model with the unified deployment path retained at inference, which is the gap UniReflow closes.

Rectified flow and distillation. Rectified flow [2] straightens learned ODE paths via reflow, in which start-end pairs are resampled along the learned flow. Conditional flow matching [8] and minibatch optimal transport [22] provide a simulation-free training objective related to optimal transport [15]. InstaFlow [3] distills Stable Diffusion to one step but requires multiple reflow iterations because noise-to-image transport is highly curved; scaling reflows to high-resolution transformers [23] further validates the approach. Progressive distillation [24], consistency models [26], and distribution matching distillation [27] accelerate diffusion models in single-task settings. We exploit paired-modality near-linearity to deliver a single direction-conditioned student after one reflow iteration.

Semantic image synthesis. Paired translation [29] and high-resolution conditional GANs [30] established layout-to-image synthesis. SPADE [7], OASIS [31], and SDM [32] condition GANs on label maps via spatially-adaptive normalisation. SC-GAN [33] exploits semantic composition. Diffusion-based methods including Palette [19] and BBDM [4] build on pretrained latent diffusion [9]

with classifier-free guidance [28]. All these methods are unidirectional: they generate images from layouts but cannot segment.

Discriminative segmentation. Atrous CNN models [6], Transformer-based segmenters [10], and universal segmentation [5,11] achieve high accuracy on specialist benchmarks but cannot synthesise. Diffusion features also yield label-efficient segmentation [12]. These task-specific designs motivate unified flow alternatives. Parameter-efficient adaptation via LoRA [13] has become a standard tool for diffusion personalisation. We extend it to flow-matching velocity fields and find that attention-only adaptation is insufficient, since convolutional layers are critical for sharp semantic boundaries.

3. METHOD

3.1. Preliminaries: flow matching

Flow matching [8] learns a time-dependent velocity field whose induced ODE transports a source distribution to a target distribution. Concretely, the ODE is

$$\frac{dz_t}{dt} = v_\theta(z_t, t), \quad t \in [0, 1]. \quad (1)$$

Under the linear interpolation path $z_t = (1 - t)x + tm$, the conditional flow-matching (CFM) objective reduces to

$$\mathcal{L}_{CFM} = \mathbb{E}_{t \sim U[0,1], (x,m) \sim \pi} [\|v_\theta(z_t, t) - (m - x)\|_2^2], \quad (2)$$

where π is the data coupling pairing each image latent x with its mask latent m . We operate in the latent space of Stable Diffusion 1.5 [9], encoding both images and label maps through its frozen VAE encoder [16]. Masks are perturbed with uniform noise of amplitude 0.1 before encoding, following [1], to prevent degenerate latents.

3.2. Bidirectional flow matching teacher

The teacher is a single SD 1.5 UNet with LoRA adapters of rank 128 applied to attention projections (q, k, v , out), feed-forward projections, and the convolutional layers conv1, conv2, and conv_shortcut inside each residual block. Around 10 percent of total parameters are trainable. The same velocity field serves both directions: forward Euler integration from $t = 0$ to $t = 1$ yields segmentation; reverse Euler integration from $t = 1$ to $t = 0$ yields synthesis, with no architectural modification. Including convolutional layers is essential: attention-only adapters fail to learn sharp semantic boundaries because the required local structural capacity is convolutional. The CFM loss in Eq. (2) is minimised with uniform timestep sampling and horizontal-flip augmentation.

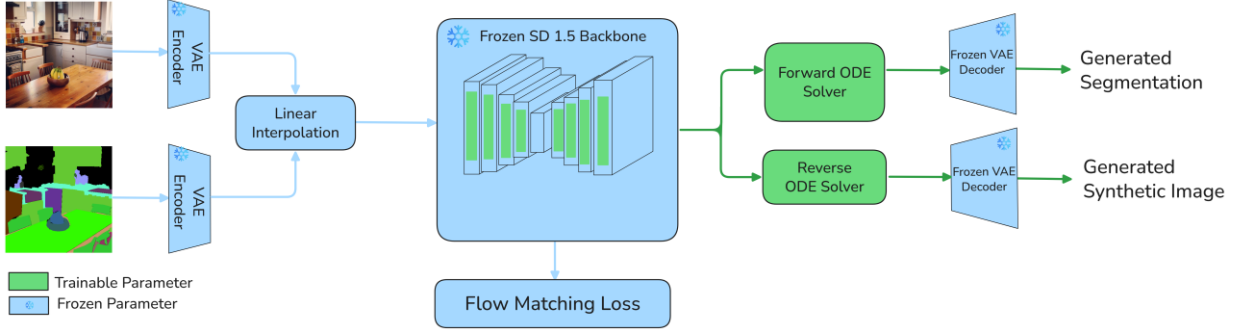


Figure 1: UniReflow pipeline. A frozen SD 1.5 backbone with LoRA adapters learns a bidirectional velocity field on image-mask latents. Reflow produces straight endpoint pairs; the one-step student adds a direction embedding to route the shared backbone to segmentation (forward ODE) or synthesis (reverse ODE) in a single Euler step.

3.3. Trajectory straightening via reflow

The teacher produces high-quality results with 25 Euler steps, but performance degrades sharply at one step because the learned ODE trajectories are curved. Reflow [2] addresses this by constructing a new coupling whose endpoints lie on approximately straight lines. For each training sample we integrate the teacher’s ODE forward (image to mask) or backward (mask to image) for 50 Euler steps and store the endpoint pair. We generate 50K pairs per direction. Because both endpoints encode the same scene, the initial coupling already approximates the optimal-transport map, and a single reflow iteration suffices. Noise-to-image transport typically needs multiple iterations [3]. Sec. 4.3 quantifies this empirically via the straightness integral S .

3.4. Direction-conditioned unified student

The core contribution of this paper is a single direction-conditioned student. We attach a learnable embedding $E_d \in \mathbb{R}^{2 \times d_t}$ to the UNet class-embedding slot, where d_t is the time-embedding dimension. A direction tensor $d \in \{0, 1\}$ (0 for SS, 1 for SIS) is added to the time embedding on every forward pass:

$$\tau(t, d) = \text{TimeEmb}(t) + E_d[d]. \quad (3)$$

This requires no architectural surgery beyond a single zero-initialised embedding: the Diffusers UNet2DConditionModel already sums `class_embedding(d)` into the time embedding when present. Both directions share this architecture and inherit the teacher weights at initialisation. We then train one LoRA adapter per direction with identical rank and the same one-step distillation objective, so the only inference-time switch is the direction tag fed into $\tau(t, d)$ together with the matching adapter.

Let v_ϕ denote the student velocity prediction at the distillation timestep and $D(\cdot)$ the frozen VAE decoder. The per-direction distillation objective is

$$\mathcal{L}_{\text{distill}} = \mathbb{E}_{SS} \|\tilde{m} - \hat{m}\|_2^2 + \mathbb{E}_{SIS} \text{LPIPS}(D(\tilde{x}), D(\hat{x})), \quad (4)$$

where $\tilde{m}, \tilde{x}$ are the student’s one-step predictions and $\hat{m}, \hat{x}$ the reflow targets; the SS adapter optimises the first term on SS reflow pairs ($d = 0$) and the SIS adapter optimises the second on SIS reflow pairs ($d = 1$). The asymmetric loss reflects the nature of each target distribution: segmentation masks are low-entropy discrete label maps for which MSE suffices, while natural images are high-entropy continuous signals where LPIPS [25] encourages sharp, perceptually coherent outputs. Both adapters are initialised from the same teacher with E_d zero-initialised, so each starts equivalent to the teacher at step 0 and the only direction-specific state is the LoRA delta. This per-direction-adapter parameterisation keeps the architectural unification and one-pass inference intact.

3.5. Why image-mask transport is near-linear

We define trajectory straightness following [2] as

$$S(v_\theta) = \mathbb{E}_{(x_0, x_1) \sim \pi} \left[\int_0^1 \|v_\theta(z_t, t) - (x_1 - x_0)\|_2^2 dt \right], \quad (5)$$

where $z_t = (1 - t)x_0 + tx_1$ is the linear interpolant; $S = 0$ if and only if every ODE trajectory is a straight line. In noise-to-image flow matching the source is unstructured Gaussian noise while the target encodes scene geometry, so the transport must reorganise spatial structure entirely and trajectories are highly curved. In image-mask transport both endpoints encode the same scene: object boundaries, layout, and semantic extent are shared, and the latents differ only in the modality of the encoded signal. The displacement $m - x$ has low spatial variance compared to a random permutation of endpoints, and the data coupling is therefore close to the Wasserstein-2 optimal-transport plan [15], implying small S .

Remark (near-linearity of image-mask ODE paths).

Let v^* be the velocity field of the conditional OT plan for the image-mask coupling. When the marginal transport cost $\mathbb{E}\|x_0 - x_1\|^2$ is near-minimal, v^* almost everywhere coincides with the linear-interpolant velocity (x_0, x_1) and $S \rightarrow 0$ [2, 15]. The learned v_θ approximates v^* under the CFM objective, so the same near-linearity transfers to the learned flow. Sec. 4.3 measures S directly and shows it is

more than an order of magnitude smaller for image-mask than for image-noise transport on the same teacher.

4. EXPERIMENTS

4.1. Setup

We evaluate on COCO-Stuff [17] (5000 validation images, 171 semantic classes) and CelebAMask-HQ [14] (2000 validation faces, 19 fine-grained classes) at 512×512 resolution. All models use SD 1.5 [9] as backbone with 4-channel VAE latents. We report mIoU for segmentation

and FID [30] and LPIPS [25] for synthesis. Two teacher variants are reported: full fine-tune of the backbone (860M parameters) and LoRA (rank 128, ~89M trainable parameters). Efficiency is measured on a single RTX 4090 in fp16. All metrics use deterministic inference on fixed validation splits. On CelebAMask-HQ SS, the one-step unified student attains **55.7 mIoU** and **90.9%** pixel accuracy on 1000 validation faces, the same one-reflow-suffices behaviour we observe on COCO-Stuff.

4.2. Comparison with prior work

Table 1: Comparison with prior work. Top: task-specific baselines. Bottom: unified methods. Arrows mark whether higher ($\uparrow$) or lower ($\downarrow$) is better; gray rows are our one-step distilled students.

Method	Task		Sampler	Steps	COCO-Stuff	CelebAMask-HQ	
	SS	SIS			mIoU $\uparrow$	FID $\downarrow$	LPIPS $\downarrow$
DeepLabV2 (TPAMI'18) [6]	✓	—	CNN	1	33.2	—	—
MaskFormer (NeurIPS'21) [11]	✓	—	Transformer	1	41.9	—	—
SegFormer (NeurIPS'21) [10]	✓	—	Transformer	1	46.7	—	—
Mask2Former (CVPR'22) [5]	✓	—	Transformer	1	53.7	—	—
SPADE (CVPR'19) [7]	—	✓	GAN	1	—	42.2	0.487
pix2pixHD (CVPR'18) [29]	—	✓	GAN	1	—	54.7	0.529
SC-GAN (ICCV'21) [33]	—	✓	GAN	1	—	19.2	0.395
BBDM (CVPR'23) [4]	—	✓	BBDM	200	—	21.4	0.370
SemFlow (NeurIPS'24) [1]	✓	✓	Euler	25	38.6	32.6	0.393
SemFlow (NeurIPS'24) [1]	✓	✓	Euler	1	28.3	—	—
UniReflow Teacher (Full FT)	✓	✓	Euler	25	40.8	27.1	0.358
UniReflow Teacher (LoRA)	✓	✓	Euler	25	34.6	32.0	0.390
UniReflow Student (Full FT)	✓	✓	Euler	1	39.2	28.3	0.379
UniReflow Student (LoRA)	✓	✓	Euler	1	33.2	31.4	0.406

Among one-step dual-task methods, our unified architecture is the strongest entry: in a single Euler step it matches the 25-step teacher on CelebAMask-HQ FID and LPIPS and improves COCO mIoU over SemFlow Euler-1 by 4.9 points at the LoRA scale and 10.9 points at the full-FT scale. Task-specific specialists still lead within their single direction (Mask2Former on SS, BBDM on SIS), but none of them can serve the opposite task; the table accordingly reports them only in the column where they operate. UniReflow occupies the dual-task one-step Pareto point that no prior unified framework reaches.

4.3. Trajectory straightness validation

Protocol. We compute S as a Monte-Carlo estimate of Eq. (5), accumulating over 25 midpoint-rule substeps along the linear interpolant $z_t = (1 - t)x_0 + tx_1$, following the standard definition [2]. We sample 5,000 latent pairs (x_0, x_1) per dataset and evaluate $\|v_\theta(z_t, t) - (x_1 - x_0)\|_2^2 dt$ with the same teacher and VAE. The paired rows set $z_1 = \text{VAE}_{enc}(\text{mask}_{rgb})$. The unpaired rows replace z_1 with an independent Gaussian sample of the same shape.

Table 2: Trajectory straightness S on the same teacher, grouped by dataset. Each row reports the unpaired (image, noise) and paired (image, mask) couplings side by side; the ratio shows how much endpoint coupling alone straightens the flow. Lower S is straighter.

Dataset	Unpaired $S\downarrow$	Paired $S\downarrow$	Ratio
COCO-Stuff	2.456	0.218	11.3×
CelebAMask-HQ	4.418	0.115	38.4×

Table 2 isolates the contribution of endpoint coupling. We evaluate one learned velocity field along two couplings on the same data, holding architecture, dataset, optimiser, and schedule fixed. The only variable is the choice of x_1 . The paired coupling reduces S by 11.3× on COCO-Stuff and 38.4× on CelebAMask-HQ, which directly measures how much endpoint coupling alone straightens the trajectory. Two further observations support the operational claim. First, the pre-reflow image-mask teacher already gives a usable 1-step solution at FID 128.79 and mIoU 15.4 (Table 3a), while pure noise-to-image collapses entirely. Second,

after a single reflow iteration the 1-step student reaches FID 56.41 and mIoU 33.2 on COCO-Stuff, within 2% relative on FID and 4% on mIoU of the same-configuration 25-step teacher (Table 3a). InstaFlow [3] requires two reflow rounds for competitive 1-step quality on noise-to-image transport, giving an indirect external corroboration.

4.4. Ablation and efficiency analysis

Table 3: Reflow ablation (a), step-quality curve (b), and inference latency (c). Δ rows give the one-step student minus its 25-step teacher; gray marks the deployed model.

(a) Reflow ablation, COCO-Stuff.

Role	Adapter	Reflow	Steps	FID↓	mIoU↑
Teacher	LoRA	0	25	57.53	34.6
Teacher	LoRA	0	1	128.79	15.4
Student	LoRA	1	1	56.41	33.2
Δ vs. 25-step teacher				-1.12	-1.4
Teacher	Full FT	0	25	42.14	40.8
Student	Full FT	1	1	46.25	39.2
Δ vs. 25-step teacher				+4.11	-1.6

(b) Step quality, LoRA teacher on COCO-Stuff SIS.

Model	Steps used	FID↓
Teacher	1	128.79
Teacher	4	105.32
Teacher	8	83.61
Teacher	25	57.53

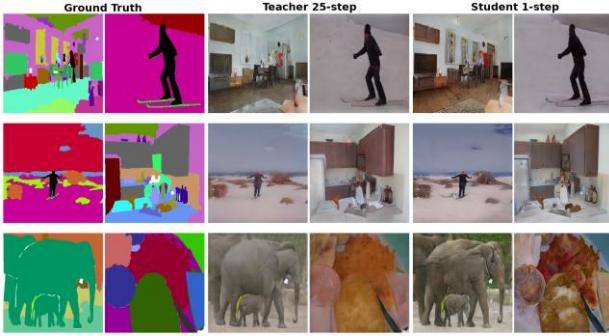


Figure 2: Qualitative SIS on COCO-Stuff. Each triple: ground-truth mask, 25-step teacher output, one-step student output (SIS-direction LoRA adapter).

The unified student preserves global structure and semantic fidelity across both directions. In synthesis, the teacher renders finer textures in high-frequency regions while the student captures composition and dominant shapes. In segmentation, the teacher better resolves thin structures that benefit from iterative boundary refinement, which is consistent with the straightness measurement of Table 2: image-mask trajectories are close to linear in the bulk but not at high-frequency boundaries. Common failure modes are consistent across both directions: hair

Model	Steps used	FID↓
Teacher	50	51.07
Student	1	56.41

(c) Inference latency, RTX 4090, 512×512, fp16.

Role	Steps	Latency (ms)↓	FPS↑	Speed-up
Teacher	25	2775	0.4	1.0×
Unified Student	1	175	5.7	15.9×

Table 3a isolates the reflow contribution. Without reflow, the LoRA teacher at 1 step produces FID 128.79 and mIoU 15.4%. The velocity field is learned correctly, but the trajectories are curved, so a single Euler step overshoots. After reflow, the student nearly matches the 25-step teacher on SIS (FID 56.41 vs. 57.53) and recovers most SS quality (33.2% vs. 34.6% mIoU). The full-fine-tune path behaves the same way: the student stays within 10% of the teacher on SIS (FID 46.25 vs. 42.14) and reaches 39.2% vs. 40.8% mIoU on SS. LoRA and full fine-tune benefit equally, so reflow effectiveness depends on transport geometry rather than model capacity. Replacing 25 Euler steps with 1 cuts wall-clock latency from 2775 to 175 ms per image (Table 3c) and the teacher’s quality survives the cut. One step is the right operating point for this transport, not a lossy approximation.

4.5. Qualitative results



Figure 3: Qualitative SS on COCO-Stuff. Each triple: input image, 25-step teacher mask, one-step student mask (SS-direction LoRA adapter). The unified student preserves semantic layout while running at 5.7× the teacher’s throughput.

strands, thin structures such as poles and fences, and small accessories collapse under one-step inference, since these high-frequency details rely on the iterative refinement of the multi-step teacher.

5. CONCLUSION

We presented UniReflow, a three-stage pipeline — flow matching, reflow, then distillation — that collapses 25-step unified perception and generation into one forward pass. What makes a single step enough is the near-linearity

of image-mask transport: the two domains share boundaries and layout, so one reflow iteration straightens the trajectories for accurate one-step Euler integration. We measure that straightness directly ($11.3\times$ to $38.4\times$ flatter than an image-noise baseline on the same teacher). The resulting direction-conditioned student runs both segmentation and synthesis in one pass, stays within 1.6 mIoU of the 25-step teacher on COCO-Stuff, and in its LoRA form fits on a consumer GPU. The broader point is a design principle: when source and target share spatial correspondence, straightening the transport is a better route to fast inference than adding model capacity or integration steps. Several directions remain open: a formal curvature metric to replace the straightness integral we report here, wider benchmarks such as ADE20K and Cityscapes at multiple resolutions, boundary-aware objectives (boundary loss, edge attention, multi-scale distillation), text-conditioned generation, and the same single-reflow recipe on other aligned-transport tasks such as depth estimation and edge-guided synthesis.

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